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A RELATED WORK

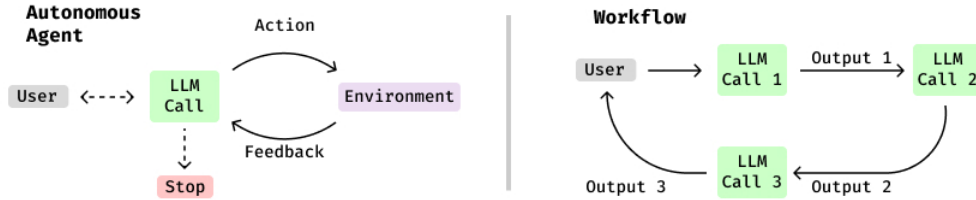


Figure 5: **Agent architectures.** Autonomous agents delegate branching decisions to the LLM, which chooses actions dynamically, while workflows follow a fixed, pre-defined sequence of calls.

Agentic Frameworks. A growing line of work designs LLM agents that interleave reasoning and action in structured loops. ReAct couples verbal “thoughts” with external actions to improve planning and tool use (Yao et al., 2023), and StateAct extends this idea with self-prompting and explicit state-tracking to reduce goal drift (Rozanov & Rei, 2024), though their benefits may partly stem from prompt artifacts rather than genuine reasoning (Verma et al., 2024). CodeAct unifies the action space by generating and executing Python code (Wang et al., 2024), while Reflexion enables agents to iteratively improve via self-feedback and reflection (Shinn et al., 2023). Larger-scale systems such as AutoGPT, BabyAGI, and DeepResearch demonstrate cycles of planning, execution, and retrieval that unlock more autonomous behavior (Sapkota et al., 2025; Huang et al., 2025b), and embodied agents like Voyager showcase continual skill learning and adaptation in open-ended environments (Wang et al., 2023).

External Memory Systems. A large body of work augments LLMs with mechanisms for storing and retrieving information beyond the context window. Retrieval-Augmented Generation (RAG) couples parametric knowledge with external retrieval (Lewis et al., 2020), with RAG+ grounding retrieval in task exemplars (Wang et al., 2025) and HippoRAG modeling hippocampal indexing for multi-hop reasoning (Gutiérrez et al., 2024). Beyond static retrieval, a distinct line of research addresses long-horizon consistency, often benchmarked on LOCOMO (Maharana et al., 2024; Parviz & Yoshida, 2025b), by treating memory as a hierarchical or operating-system-level challenge. Pioneering this approach, MemGPT introduces virtual context management to support unbounded contexts (Packer et al., 2023), while MemoryOS implements a hierarchical storage architecture (short, mid, and long-term) with dynamic updating strategies (Kang et al., 2025). Similarly, LightMem adopts a three-stage Atkinson-Shiffrin model for efficient retrieval (Fang et al., 2025), and A-mem utilizes Zettelkasten-style networking for evolving knowledge (Xu et al., 2025). Further optimizations include Mem0, which extracts salient facts for personalization (Chhikara et al., 2025), and Memory-R1, which leverages reinforcement learning for structured memory operations (Yan et al., 2025). In parallel, other systems target behavioral simulation, such as Generative Agents (Park et al., 2023) and Reflexion (Shinn et al., 2023), while frameworks like LangChain, LlamaIndex, and CrewAI provide tooling for these architectures (Chase et al., 2022; Liu et al., 2023; Moura et al., 2024). However, while these methods address factual recall and long-term coherence, none enable agents to generate and preserve hidden, task-specific private state across turns, a capability needed to solve PSITs.

Cognitive Architectures. Several frameworks draw on cognitive science to structure memory and reasoning. HIA-GENT uses hierarchical working-memory management with chunking (Hu et al., 2024), CoALA integrates episodic, semantic, and procedural memory within a structured decision cycle (Sumers et al., 2023), MemOS treats memory as a schedulable resource (Li et al., 2025), and the Unified Mind Model proposes a layered architecture with a central working-memory core (Hu & Ying, 2025). Other work emulates cognitive operations through prompting: Cognitive Prompting enforces structured steps like goal clarification and integration (Kramer & Baumann, 2024), and Cognitive Tools equip LLMs with modular reasoning primitives (Ebouky et al., 2025; Sghaier et al., 2026). While these approaches highlight the value of modular design and structured reasoning, they do not support maintaining and acting upon private hidden state across interactions, and thus do not address PSITs.

Game-Based and Memory Benchmarks. Recent research has increasingly utilized interactive games and specialized benchmarks to evaluate agentic capabilities. Karabag et al. (2025); Jha et al. (2026) investigate information control in “The Chameleon” game, analyzing whether agents can strategically conceal information; however, in their setting, the secret is an assigned input rather than a value generated by the agent itself. Similarly, Hu et al. (2025) introduce MemoryAgentBench to evaluate core competencies such as retrieval and selective forgetting, yet they focus on the

retention of external information accumulating over turns rather than the maintenance of internal working memory. In the domain of role-playing, Peng & Shang (2025) propose codified profiles to enforce behavioral consistency via executable functions. While these works advance the evaluation of agentic memory and consistency, they rely on pre-defined secrets, externalized logic, or the recall of observation history. In contrast, PSITs specifically challenge the agent to generate a private state and maintain it internally, a distinct cognitive loop that remains under-explored.

Faithfulness and Self-Consistency. Self-consistency checks are a common tool for assessing faithfulness in interpretability research (Parcalabescu & Frank, 2024; Madsen et al., 2024; Mayne et al., 2025; Sghaier et al., 2026). Our work can be seen in this light, as we test whether an agent’s hidden state aligns with its public behavior. Importantly, models rarely admit upfront that they cannot play Hangman without private memory, since, as highlighted by Kalai et al. (2025), training directly incentivizes to bluff rather than abstain.

B MEMORY REPRESENTATION

To help agents use the working memory effectively and write down relevant yet concise items, we drew inspiration from cognitive models of working memory as well as recent work on language-agent architectures. Classic cognitive theories highlight the modularity of working memory, most notably Baddeley’s multi-component model with its central executive and episodic buffer (Baddeley, 1983), while HiAgent demonstrates the value of hierarchical goal structuring for long-horizon tasks (Hu et al., 2024). The CoALA framework further emphasizes the role of modular reasoning outputs that can be stored and acted upon in subsequent decision cycles (Sumers et al., 2023). Finally, recent work on cognitive prompting (Kramer & Baumann, 2024) and cognitive tools (Ebouky et al., 2025) shows that guiding LLMs with cognitively inspired structures improves reasoning and task performance, motivating our design choices.

Accordingly, we structure the private working memory into three sections:

- **Goals / Plans.** Inspired by the central executive in Baddeley’s model and hierarchical planning in HiAgent, this section records the agent’s current objectives and strategies.
- **Facts / Knowledge / Episodic buffer.** Following Baddeley’s episodic buffer, this section stores salient facts and contextual knowledge extracted during the interaction.
- **Inference / Reasoning / Active Knowledge.** In line with CoALA, this section captures reasoning outputs and intermediate inferences that the agent intends to act upon in the next decision cycle.

The agent is explicitly instructed within the system prompt on the semantic purpose of each section and is tasked with maintaining them throughout the interaction. The complete prompts and instructions are provided in Appendix G.

C QUALITATIVE ANALYSIS OF PRIVATE STATES

To provide insight into the performance differences reported in Section 5, we present here a qualitative comparison of the internal states maintained by different agent architectures.

The examples below capture the agent’s private context snapshot immediately prior to the forking point (t_{fork}).

- For Private Working Memory Agents (Workflow/Autonomous), we display the explicit text block currently stored in their private memory variable.
- For Private CoT, we show the most recent segment of the accumulated reasoning history.
- For Memory-Augmented Baselines (Mem0, A-Mem, LightMem), we display the specific memory chunks retrieved.

These snapshots illustrate the fact that statefulness differs from context retrieval. While successful agents possess a memory block that explicitly and unambiguously states the secret, retrieval baselines often surface semantically relevant but state-irrelevant details (e.g., rules of the game or past public utterances), leaving the model effectively amnesic regarding the hidden variable.

The following examples are extracted from runs using GPT-OSS 120B on the Hangman task.

Autonomous Agent with Overwrite Memory Tool

- ## 1. Goals and Plans
- Host Hangman game: secret word chosen, track lives, guesses.
- ## 2. Facts and Knowledge
- Secret word: <secret>planet</secret>
 - Word length: 6
 - Remaining lives: 5
 - Guessed letters: ["d", "a", "t"]
 - Current pattern: __ a __ t
- ## 3. Active Notes
- User guessed "d" which is not in the word.
 - User guessed "a" which is in the secret word at position 3.
 - User guessed "t" which is in the secret word at position 6.
 - Awaiting next guess.

Workflow Agent with Overwrite Memory Tool

- ## 1. Goals and Plans
- Host Hangman game: maintain secret word, track guesses, lives (6), update pattern after each guess, respond with pattern, lives, guessed letters.
- ## 2. Facts and Knowledge
- <secret>puzzle</secret>
- ## 3. Active Notes
- Pattern: _ _ _ _ _
- Lives remaining: 3
- Guessed letters: [s, r, a]

Mem0

- Wants to guess letter "h" @ 2025-10-20T11:30:48.209999-07:00
- Requests secret word to be an existing English word in lowercase @ 2025-10-20T11:30:48.231146-07:00
- Wants to guess letter "b" @ 2025-10-20T11:30:22.019822-07:00
- Wants to guess letter "i" @ 2025-10-20T11:30:48.220349-07:00
- Specifies 6 lives for the Hangman game @ 2025-10-20T11:30:22.008008-07:00
- Wants to play Hangman with the assistant as host @ 2025-10-20T11:30:21.983470-07:00

A-mem

- Content: My next guess is the letter "e". Is it in the secret word?
Keywords: letter, guess, secret word
Tags: -
Context: General
 - Content: My next guess is the letter "o". Is it in the secret word?
Keywords: guess, letter, "o", secret word
Tags: -
Context: General
 - Content: My next guess is the letter "a". Is it in the secret word?
Keywords: letter, guess, secret word, a
Tags: -
Context: Let's play Hangman. You will be the host.
- Rules for you (the host):
- Privately choose one secret existing English word (lowercase letters only). Do not reveal it unless I ask to reveal it.